

**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI**

First Semester 2026-2027

**CHE F313 Separation Processes-II**

**Quiz - 1 (Closed Book)**

**Duration: 15 Mins**

**Date: 03/09/2026**

**Marks: 10**

Name: Sample Solution ID No.:                     

A binary solid mixture consists of Component A and Component B. The overall composition of A in the mixture is 50% by mass. Samples are withdrawn from different locations after mixing. Each sample has a total mass of 100 g, and the mass of Component A in each sample is measured as follows:

| Sample | Mass of A (g) | Sample | Mass of A (g) | Sample | Mass of A (g) | Sample | Mass of A (g) |
|--------|---------------|--------|---------------|--------|---------------|--------|---------------|
| 1      | 52            | 4      | 45            | 7      | 54            | 10     | 50            |
| 2      | 48            | 5      | 51            | 8      | 46            |        |               |
| 3      | 55            | 6      | 49            | 9      | 50            |        |               |

Assuming each 100-g sample contains 1000 particles. Calculate the mixing index.

$$\text{Mixing Index, } I_s = \frac{\sigma_0}{S}$$

$$\sigma_0 = \sqrt{\mu(1-\mu)}$$

$$S = \sqrt{\frac{\sum_{i=1}^N x_i^2 - \bar{x} \sum_{i=1}^N x_i}{N-1}}$$

$$\mu = 0.5 \quad (\text{given})$$

$$\sigma_0 = \sqrt{0.5(1-0.5)}$$

$$\boxed{\sigma_0 = 0.5}$$

where,

$x_i$  = Mass fraction of A in each sample

$\bar{x}$  = Avg. value of mass fraction

$N$  = Number of Spot samples

Total mass = 100g

| Sample | Mass of A (g) | Mass fraction ( $x_i$ ) | $x_i^2$                       | $\bar{x}$ |
|--------|---------------|-------------------------|-------------------------------|-----------|
| 1.     | 52            | 0.52                    | 0.2704                        | 0.5       |
| 2.     | 48            | 0.48                    | 0.2304                        |           |
| 3.     | 55            | 0.55                    | 0.3025                        |           |
| 4.     | 45            | 0.45                    | 0.2025                        |           |
| 5.     | 51            | 0.51                    | 0.2601                        |           |
| 6.     | 49            | 0.49                    | 0.2401                        |           |
| 7.     | 54            | 0.54                    | 0.2916                        |           |
| 8.     | 46            | 0.46                    | 0.2116                        |           |
| 9.     | 50            | 0.50                    | 0.25                          |           |
| 10.    | 50            | 0.50                    | 0.25                          |           |
|        |               | $\sum_{i=1}^N x_i = 5$  | $\sum_{i=1}^N x_i^2 = 2.5092$ |           |

$$S = \sqrt{\frac{\sum_{i=1}^N x_i^2 - \bar{x} \sum_{i=1}^N x_i}{N-1}}$$

$$= \sqrt{\frac{2.5092 - (0.5 \times 5)}{10-1}}$$

$$= 0.031$$

$$I_s = \frac{\sigma_0}{S} = \frac{0.5}{0.031}$$

$$= 16.12$$

$$I_s = 16.12$$

→ mixing index =